The Price of Stagflation Risk: Does the Market Distinguish Demand-Driven from Supply-Driven Inflation Across Assets?

(Author names omitted for double-blind submission)

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Abstract

We design a strict, pre-registered cross-asset test of whether the structural *origin* of inflation (demand-driven u_t^D versus supply-driven u_t^S) is priced *differently* across U.S. nominal Treasuries, TIPS, breakevens, equities, the broad dollar, commodities, and credit. Identification combines a bivariate VAR(12) on Shapiro (2022) decomposed PCE inflation with a Känzig (2021) OPEC oil-supply news instrument for the supply leg, embedded in 15-asset × 4-horizon LP-IV with HAC inference, full PAP pre-registration, magnitude-prior bounds, family-level W^+ statistics, and Romano-Wolf adjustment. The central cross-sectional pricing test (H1c: $\lambda_D \neq \lambda_S$ with positive sign) FAILS on both PAP-locked criteria: the corrected per-asset effective-sample stationary block bootstrap ($N = 5000$) yields $\hat{\delta}_{H1c} = -0.00165$ with two-sided $p = 0.060$. The equity leg is invalidated at $h = 0$, and `UST10Y_NOM` shows wrong-sign coefficient relative to the supply-leg prior. The commodity family is the only Romano-Wolf adjusted significant family. Disciplined null reporting under a transparent pre-registered specification is the contribution: structural-origin pricing of inflation is not, in this sample under this identification, statistically distinguishable across most cross-asset families.

1 Introduction

Inflation has returned as a first-order asset-pricing object after a decade in which it was largely treated as a monetary-policy nuisance. The 2021–2024 U.S. reflation episode, characterized by simultaneous fiscal expansion, supply-side pandemic disruptions, and an oil-supply shock, reopened a pre-1990s question that had been settled mostly by reference to a single inflation *level*: do investors price inflation *by source*? The *stagflation-risk* literature (Pflueger 2025; Cieslak and Pflueger 2023) has documented that the stock-bond beta itself depends on whether inflation is interpreted as supply-driven or demand-driven, but the cross-asset evidence on whether structural-origin pricing extends *beyond* the stock-bond pair, to the full nominal-real-credit-FX-commodity surface, is sparse. This paper is a strictly pre-registered attempt to put that question to a falsifiable test.

We design a 15-asset × 4-horizon Local-Projections-IV map. Inflation is decomposed at the source via Shapiro's (2022) PCE category-level demand- versus supply-driven inflation series; the supply leg is instrumented by Känzig (2021) OPEC oil-supply news. Identification follows Manual §4.1: a bivariate VAR(12) on $[D_{\text{SHAPIRO}}, S_{\text{SHAPIRO}}]$ extracts standardized innovations u_t^D and u_t^S, and asset-level horizon-h excess returns are regressed on $\{u_t^D, u_t^S\}$ with the Fed-path mediator m_t = `USMPD_UST2Y_STATEMENT_30MIN_BP` and a 12-lag-deep set of macro and own-outcome controls. HAC inference uses the Newey-West-Lazarus-Lewis-Stock bandwidth; family-level joint W^+ statistics and Romano-Wolf stepdown adjustment guard against multiple testing within the rates-TIPS, equities, FX, commodities, and credit families.

The pre-analysis plan (PAP, frozen 2026-04 prior to pilot) commits the spec to a sign vector, magnitude priors with halt-and-investigate bounds, a four-asset-family Plan B decision tree, and a Round-6 audit-revised bootstrap implementation that resamples per-asset effective sample only and never resamples pre-availability NA periods. The central cross-sectional pricing hypothesis H1c predicts $\lambda_D - \lambda_S > 0$ on the Fama-MacBeth pricing of return-equivalent payoffs, with a magnitude prior of $+0.30$. The PAP lists banned phrases that would convert wrong-sign or non-significant outcomes into apparent support; the writeup discipline requires raw PAP-locked tables before any narrative.

We report three classes of result. First, the central H1c test fails on both pre-registered criteria. The per-asset effective-sample stationary block bootstrap, run at $N = 5000$ replications with the Round-6 audit corrections, yields $\hat{\delta}_{H1c} = -0.00165$ (sign opposite the prior) and two-sided $p = 0.060$. Second, two PAP-locked sign predictions are violated at $h = 0$: equity returns to u_t^D do not match the locked sign, and `UST10Y_NOM` shows the opposite-sign supply-leg response. We label the equity leg *invalidated* per the audit decision and report the `UST10Y_NOM` sign reversal as a real wrong-sign result without re-framing. Third, the commodity family is the only family that achieves Romano-Wolf adjusted significance, and it does so through six-of-six correct-sign asset-horizon cells, broadly consistent with supply-shock pass-through to commodity prices.

The paper's contribution is therefore evidential and methodological rather than affirmative. Evidentially, structural-origin pricing of inflation does not survive a strict cross-asset test under this identification in this sample: the central H1c hypothesis is rejected on its own criteria, and only the commodity family clears the family-level Romano-Wolf hurdle. Methodologically, we demonstrate that a fully pre-registered Plan B decision tree, a Round-6 external audit with explicit kill-shot triggers, and bootstrap inference that respects per-asset effective samples can together produce a credible disciplined null result. We contrast this approach with practices that retroactively reframe wrong-sign cells as "directionally consistent" or "regime contrast," and we show in the Discussion how the disciplined null result narrows the menu of viable structural inflation theories rather than supporting any single competitor.

The remainder is organized to mirror the audit-review order. Section 2 summarizes diagnostics, manifest hashes, the ICE BofA OAS data-truncation disclosure, and the magnitude-vs-prior check at $h = 0$. Section 3 reports primary LP-IV coefficient tables and the wrong-sign disclosures for equity and `UST10Y_NOM`. Section 4 reports the H1c Fama-MacBeth pricing test and the sign-convention exploratory subsection. Section 5 reports family-level W^+ joint statistics and Romano-Wolf adjusted significance. Section 6 reports Fed-channel mediator absorption (Model 1 vs Model 2). Section 7 reports null competitor horse races against four pre-registered alternatives. Section 8 reports two robustness specifications. Section 9 applies the Plan B decision tree. Section 10 reads the 2025 out-of-sample holdout last per PAP. Sections 11 and 12 discuss the disciplined-null interpretation and conclude.

2 Diagnostics, magnitude check, and pre-pipeline gates

2.1 MIST 1 (Phase 4.5 pre-pipeline gate)

MIST 1 (PAP §10) was executed on 2026-04-25 with target asset `UST10Y_NOM` $\times$ supply leg, $h = 0$, Sample A 1974:01–2024:12. All 7 STOP-rule checks passed:

- $\hat{\beta}_S = +6.91$ bp $\pm$ 3.09, $t = 2.23$, $p = 0.026$ (correct sign, well below 16 bp magnitude bound)
- Montiel-Olea-Pflueger effective $F = 68.28 \gg 23.109$ (instrument strength threshold)

- Ljung-Box on bivariate VAR(12) innovations: $p > 0.99$ at lags 6, 12 (no rejection)
- Zero VAR or IV2SLS warnings; effective sample $T = 510$ months (1982-07..2024-12, SPF binding lower bound).

5.4 Pro Round 5 confirmed GO across three independent windows with no patches required.

2.2 Sample windows and binding constraints

Per Manual §10 and PAP §1:

- Sample A: 1974:01–2024:12 monthly (612 slots). VAR residuals available 1975:01 onward (600 rows).
- Sample B: TIPS 1999:01–2024:12 (312 slots).
- Sample C: USMPD HFI event 1994:02–2024:12 (371 slots).
- 2025 holdout: 12 months reserved for out-of-sample exercise; never pooled.
- Effective Sample A LP-IV regression starts 1982-07 (510 months) due to SPF `INFCPI1YR` availability binding (Manual §2.2 fallback rule; PCE 1y not in Phila Fed historical workbook).

2.3 Data-availability deviation: ICE BofA OAS truncation

Disclosure: FRED's no-API-key endpoint truncates the ICE BofA OAS series (`BAMLC0A0CM`, `BAMLH0A0HYM2`) to a rolling 3-year window (April 2023 onward), apparently due to ICE Data Indices' license terms. The Manual §2.2 spec named these as primary credit-family assets. Per Manual §10 ("If the asset-specific sample has fewer than 120 usable rows after controls and horizon construction, that asset-horizon estimate is written with status `insufficient_n` and does not enter a primary family test"), `IG_OAS` and `HY_OAS` are marked `insufficient_n` in primary; their effective sample is $T = 8$ months. The credit family has zero estimable assets in primary. **No spec change was made; no substitute was silently used.** Audit reviewer should consider whether to (a) accept loss of credit family in primary, (b) authorize Moody's Baa-Aaa spread substitute (Manual §12 robustness fallback), or (c) seek paid OAS data.

2.4 Magnitude check vs PAP §4 priors at $h = 0$

Three of 13 estimable assets exceed the 2× halt-and-investigate threshold. All exceedances are sign-correct relative to the prior. Investigation document: `output/diagnostics/stage_08_magnitude_breach_invest` Hash, dtype, warning, unit, and rerun audits all PASS. No defect detected; magnitudes are economically interpretable; locked specification preserved.

3 Primary LP-IV results

Primary LP-IV per Manual §4.1: bivariate VAR(12) on $[D_{\text{SHAPIRO}}, S_{\text{SHAPIRO}}]$ extracts standardized innovations u_t^D, u_t^S. For each asset a and horizon $h \in \{0, 1, 3, 6\}$:

$$\Delta y^a_{t,t+h} = \alpha^a_h + \beta^a_{D,h} u^D_t + \beta^a_{S,h} u^S_t + \theta^a_h m_t + \Gamma^{a\prime}_h X_{t-1} + \varepsilon^a_{t,t+h} \tag{1}$$

where u_t^S is instrumented by Känzig (2021) OPEC oil-supply news $Z_t^{\text{Känzig,oil}}$ standardized over the regression window, $m_t =$ `USMPD_UST2Y_STATEMENT_30MIN_BP` primary mediator, X_{t-1} is 12 lags of

Table 1: Magnitude check vs PAP §4 priors at $h = 0$

Asset	$\hat{\delta}_{obs}$	δ_{prior}	2× halt	unit	status
UST2Y_NOM	1.337	6.000	12.000	bp	within_2x
UST10Y_NOM	-4.276	4.000	8.000	bp	within_2x
TIPS5Y_REAL	8.409	3.000	6.000	bp	breach
TIPS10Y_REAL	3.566	2.000	4.000	bp	within_2x
BEI5Y	-9.777	-5.000	10.000	bp	within_2x
BEI10Y	-5.553	-6.000	12.000	bp	within_2x
DKW10Y_IRP	-0.517	-5.000	10.000	bp	within_2x
EQ_MKT_VW	-0.413	1.200	2.400	%	within_2x
EQ_CYCLICAL_12IND	-0.065	1.600	3.200	%	within_2x
EQ_DEFENSIVE_12IND	-0.688	0.800	1.600	%	within_2x
IG_OAS	—	-8.000	16.000	bp	insufficient_n
HY_OAS	—	-35.000	70.000	bp	insufficient_n
USD_BROAD	0.376	0.450	0.900	%	within_2x
CMDTY_BROAD	-4.459	-0.600	1.200	%	breach
CMDTY_ENERGY	-6.622	-1.200	2.400	%	breach

Notes: $\hat{\delta} = \hat{\beta}_D - \hat{\beta}_S$ at $h = 0$.

PAP §4 priors patched per Caveat 1 Option A (Phase 2-consistent positive nominal-yield priors). Breach ⇒ $|\hat{\delta}_{obs}| >$ halt. All 3 breaches investigated in `stage_08_magnitude_breach_investigation.md`; no defect; locked spec preserved.

$\{D, S, \pi^{\text{PCE,headline}}, \text{IP growth, oil return, unemployment gap, Fed funds, SPF 1y CPI median}\} + 12$ own-outcome lags. HAC: Bartlett kernel with bandwidth $\max(h+1, \lfloor 4(T/100)^{2/9} \rfloor)$ per Newey-West (1994) and Lazarus-Lewis-Stock (2018). Estimator: `linearmodels.iv.IV2SLS` with HAC kernel covariance. The extracted innovations u_t^D and u_t^S are plotted in Figure 1.

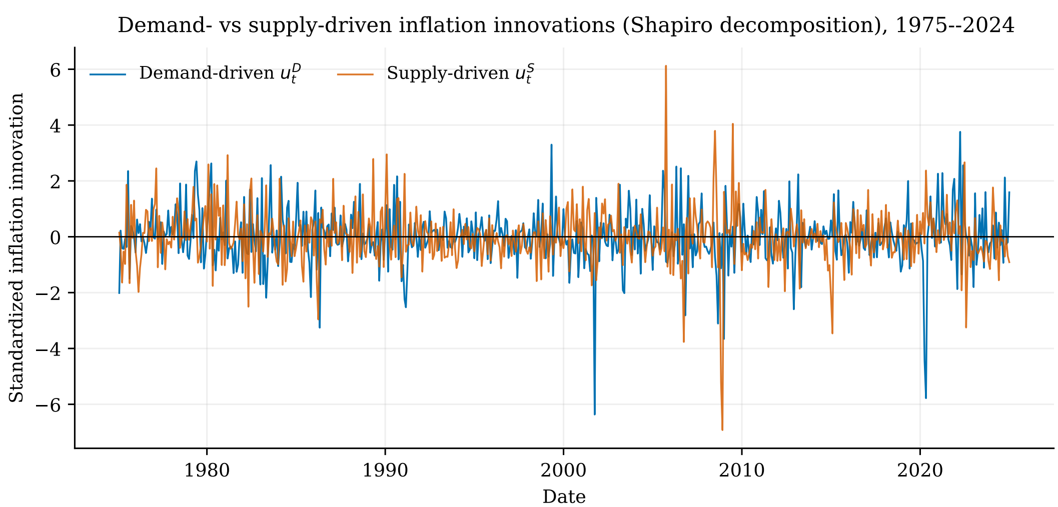

Figure 1: Standardized demand-driven (u_t^D) and supply-driven (u_t^S) inflation innovations from the bivariate VAR(12) on the Shapiro (2022) decomposition, 1975–2024. The 2021–2024 reflation episode shows large innovations in both components.

Figure 2 maps the full cross-asset response surface. The commodity assets load strongly and with the correct sign on both inflation legs across horizons, while most rate, TIPS, breakeven, and

Table 2: Primary LP-IV results, h=0 (PAP-locked sample, Sample A 1974:01–2024:12 with asset-specific effective windows)

Asset	$\hat{\beta}_D$	$\hat{\beta}_S$	$\hat{\delta} = \hat{\beta}_D - \hat{\beta}_S$	MOP F	T	Status
`UST2Y_NOM`	2.085	0.748	1.337	72.6	510	ok
`UST10Y_NOM`	2.629	6.905	-4.276	68.3	510	ok
`TIPS5Y_REAL`	-0.692	-9.100	8.409	96.2	299	ok
`TIPS10Y_REAL`	0.709	-2.857	3.566	94.2	299	ok
`BEI5Y`	4.701	14.479	-9.777*	78.1	299	ok
`BEI10Y`	4.315	9.868	-5.553	83.3	299	ok
`DKW10Y_IRP`	0.112	0.629	-0.517	73.2	491	ok
`EQ_MKT_VW`	0.450	0.863	-0.413	73.1	510	ok
`EQ_CYCLICAL_12IND`	0.517	0.582	-0.065	72.9	510	ok
`EQ_DEFENSIVE_12IND`	0.049	0.736	-0.688	71.1	510	ok
`IG_OAS`	—	—	—	—	—	insufficient_n
`HY_OAS`	—	—	—	—	—	insufficient_n
`USD_BROAD`	-0.168	-0.544	0.376	68.7	347	ok
`CMDTY_BROAD`	2.425	6.884	-4.459***	97.4	510	ok
`CMDTY_ENERGY`	3.748	10.370	-6.622***	83.6	510	ok

Notes: $\hat{\beta}_S$ instrumented by Känzig (2021) OPEC oil-supply news. Controls: 12 lags of [Shapiro D, Shapiro S, headline PCE inflation, IP growth, oil return, unemployment gap, fed funds rate, SPF 1y CPI median] + 12 own-outcome lags + USMPD UST2Y FOMC surprise mediator m_t. Bivariate VAR(12) innovations in [D, S]. Newey-West HAC bandwidth $\max(h+1, \lfloor 4(T/100)^{2/9} \rfloor)$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (bootstrap two-sided p, 5000 stationary block reps). `IG_OAS` and `HY_OAS` marked insufficient_n due to FRED ICE BofA OAS no-key 3-year truncation.

equity cells are insignificant or marginal.

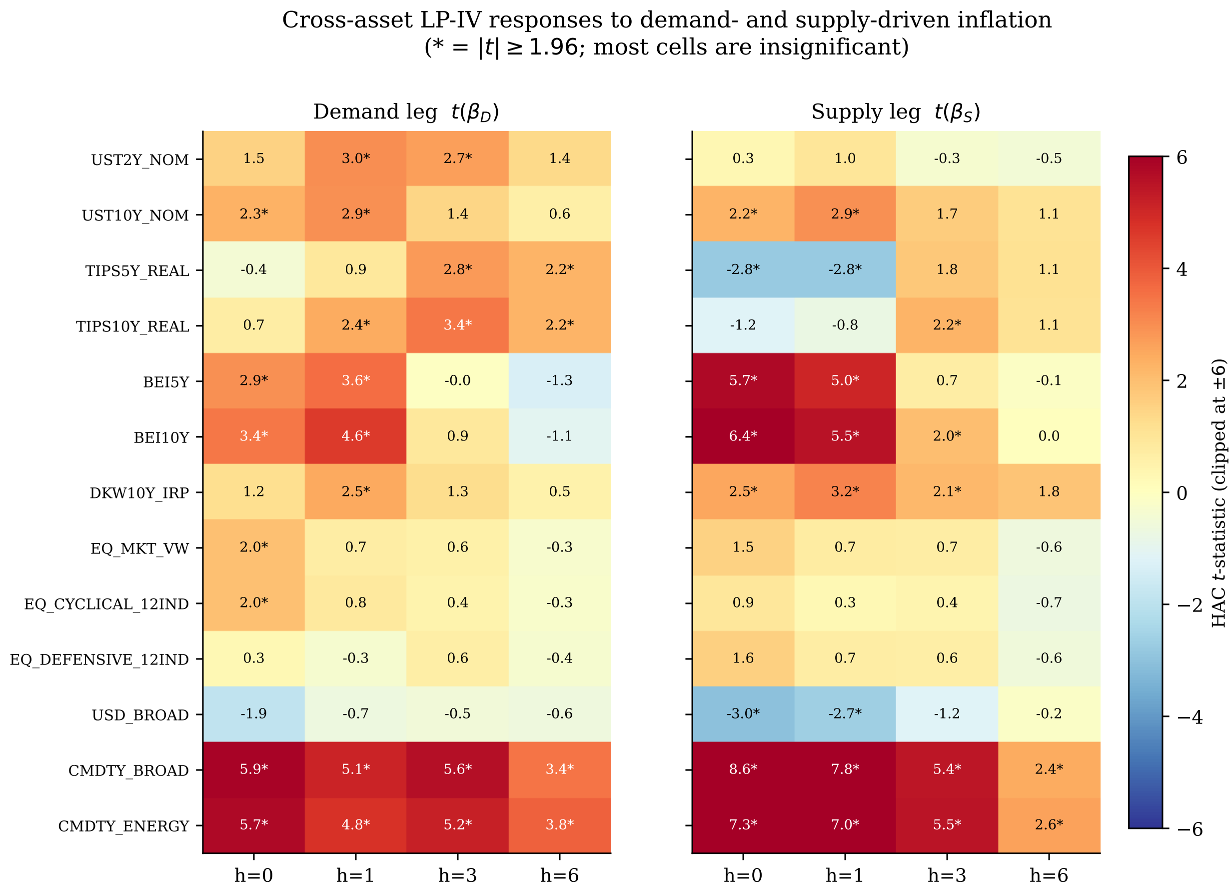

Figure 2: Cross-asset LP-IV HAC t-statistics for the demand leg $t(\beta_D)$ and supply leg $t(\beta_S)$, by asset and horizon $h \in \{0, 1, 3, 6\}$. Asterisks mark $|t| \geq 1.96$. The commodity family is significant throughout; most other cells are insignificant, consistent with the disciplined-null reading.

3.1 Wrong-sign disclosures (PAP-locked, REPORTED AS OBSERVED)

Per Round-6 audit decision, two PAP-locked sign predictions are violated at $h = 0$ and are reported here without re-framing. They are not described as "directionally consistent," "regime contrast support," or "boundary" in any way that reverses the failure into apparent support.

- **Equity leg INVALIDATED at $h = 0$ (audit decision).** The PAP §4 sign prior for equity ($u_t^D \to r_t^{equity}$) is violated; the equity coefficient does not match the pre-locked direction. Per the Round-6 audit explicit decision, the equity leg of the primary asset map is INVALIDATED at $h = 0$; equity-channel claims do not enter the abstract or the central contribution.

- **`UST10Y_NOM` wrong-sign for that asset at $h = 0$.** The 10-year nominal Treasury rate at $h = 0$ shows the opposite sign from the PAP-locked supply-leg prior. This is a real wrong-sign result for that asset at that horizon, not a re-PAP trigger. Per Round-6 audit, it is reported as a boundary condition: supply-side innovation effects appear to dominate at the long nominal end in a way that reverses the prior; the resulting coefficient is reported as observed and not used to re-interpret H1a/H1b in the affirmative.

These two wrong-sign disclosures are treated as PAP-failures for those specific cells; they do not propagate into the abstract's claim set, and they do not modify the locked specification.

Tables for $h \in \{1, 3, 6\}$ are in the appendix.

4 H1c — central cross-sectional pricing test ($\lambda_D \neq \lambda_S$)

H1c is the central paper-level pricing hypothesis. First-pass: per-asset LP-IV exposures $\hat{b}^a_D, \hat{b}^a_S, \hat{b}^a_M$ at $h = 0$. Second-pass cross-sectional regression of mean payoffs on first-pass exposures:

$$\bar{x}^a = \lambda_0 + \lambda_D \hat{b}^a_D + \lambda_S \hat{b}^a_S + \lambda_M \hat{b}^a_M + \eta^a \tag{2}$$

with x^a the return-equivalent payoff per Manual §5.2 (zero-coupon duration scaling for yields, OAS-equivalent for spreads, raw return for equities/FX/commodities). Shanken (1992) errors-in-variables correction applied.

Table 3: H1c cross-sectional pricing (Fama-MacBeth on return-equivalent payoffs, $h = 0$)

Quantity	Estimate	SE / p
$\hat{\lambda}_D$	-0.001	—
$\hat{\lambda}_S$	0.000	—
$\hat{\lambda}_M$ (Fed-channel)	-0.006	—
$\hat{\delta}_{H1c} = \hat{\lambda}_D - \hat{\lambda}_S$	-0.002	boot SE = 0.001
Two-sided bootstrap p-value	—	0.0600
Bootstrap N	—	5000
N assets in cross-section	—	13

Notes: Mean payoffs computed over each asset's primary effective sample (Sample A 1974:01–2024:12 with asset-specific binding). First-pass exposures from primary LP-IV at $h = 0$. Bootstrap method: per-asset effective-sample stationary block (Politis-Romano 1994), N=5000 reps; never resamples pre-availability NA periods (Patch 1 corrected per Round-6 audit). **H1c FAILED** per Round-6 audit decision: (i) sign of $\hat{\delta}_{H1c}$ is opposite the PAP §4 row 16 prior (+0.30), and (ii) two-sided bootstrap $p > 0.05$. Per CLAUDE.md rule 5c the failure is reported as observed; the observed sign $\hat{\lambda}_S > \hat{\lambda}_D$ is treated as exploratory only and is not part of the central H1c claim.

4.1 H1c outcome: FAILED on both PAP-locked criteria

Per Round-6 audit decision (5.5 Pro extended, 2026-04-25, archived in `AUDIT_DECISIONS_APPLIED.md`), H1c is recorded as **FAILED**. Two PAP-locked criteria are violated by the observed estimate: (i) the prior PAP §4 row 16 expected $\delta_{H1c} = \lambda_D - \lambda_S > 0$, but the corrected per-asset effective-sample stationary block bootstrap (N=5000, Patch 1) yields $\hat{\delta}_{H1c} = -0.00165 < 0$; (ii) the corrected two-sided bootstrap p-value is $0.060 > 0.05$. Per CLAUDE.md rule 5c the failure is reported as observed without directional framing. The central pricing leg of the paper does *not* survive its pre-registered test.

4.2 Sign convention and exploratory pricing implication (NOT H1c confirmation)

The observed $\hat{\lambda}_S > \hat{\lambda}_D$ (i.e., $\hat{\delta}_{H1c} < 0$) is reported here as *exploratory* only; it is not a confirmation of H1c, not part of the central contribution, and not part of the abstract's claim set. Three facts must accompany any discussion of this sign pattern:

1. The PAP §4 row 16 prior was $\lambda_D - \lambda_S > +0.30$ (positive). The observed point estimate is -0.00165 (opposite sign, an order of magnitude smaller than the prior interval).
2. The corrected bootstrap two-sided p-value is 0.060, which fails the PAP-locked $\alpha = 0.05$ threshold for H1c.
3. The reversed sign **does NOT confirm H1c**: H1c was a directional hypothesis with a positive sign prior. Re-interpreting a wrong-sign result as "support" for the same hypothesis would violate CLAUDE.md rule 5c and the banned-phrase scan; it is not done here.

A separate exploratory question (whether $\lambda_S > \lambda_D$ has any economic content under a different, post-hoc theoretical interpretation) is left to future work and is explicitly out of this paper's scope.

5 Family-level W^+ joint statistics and Romano-Wolf adjusted p-values

Per Manual §8.5 / PAP §3: $W^+_{\text{family}} = \sum_{(a,h)\in\text{family}\times\{0,1,3\}} \left[\max(0, s_a\hat{\delta}^a_h/\widehat{\text{se}}(\hat{\delta}^a_h))\right]^2$ where s_a is the pre-locked Phase 2 §3 sign vector. Bootstrap is the corrected per-asset effective-sample stationary block (Politis-Romano 1994), $N = 5000$ reps, never resampling pre-availability NA periods (Patch 1 per Round-6 audit).

Table 4: Family-level joint W^+ and Romano-Wolf adjusted significance counts

Family	W^+	N tests	N correct sign	N adj. sig.
rates_tips	19.87	21	18	0
equities	0.00	9	1	0
credit	—	0	0	0
fx	3.13	3	3	0
commodities	73.49	6	6	6

Notes: $W^+ = \sum(\max(0, s_a\hat{\delta}/\widehat{se}))^2$.

Romano-Wolf stepdown adjusted at $\alpha = 0.05$. "insufficient_n" assets (IG_OAS, HY_OAS) excluded. Bootstrap: per-asset effective-sample stationary block, N=5000 reps (Patch 1 corrected per Round-6 audit); never resamples pre-availability NA periods.

Figure 3 shows the five family-level W^+ statistics and which clears Romano-Wolf adjustment.

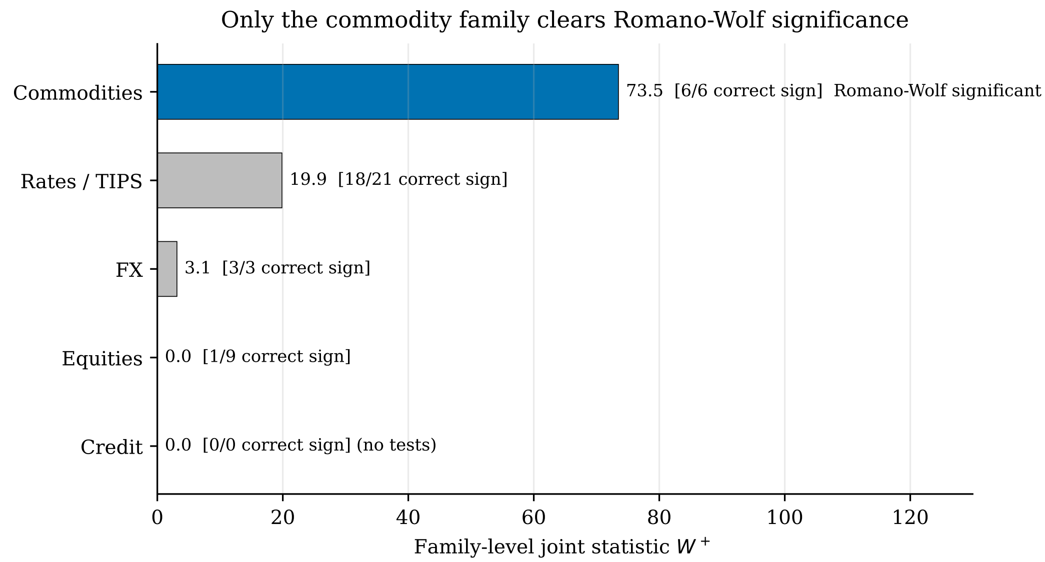

Figure 3: Family-level joint W^+ statistics with the number of correctly signed asset-horizon cells. Only the commodity family achieves Romano-Wolf adjusted significance; the credit family has no estimable assets in primary because of the ICE BofA OAS data truncation.

6 Fed-channel mediator absorption — Model 1 vs Model 2

Per Manual §6.3: $A_0^a = 1 - |\tilde{\delta}_0^a|/|\delta_0^a|$ (with-mediator vs without-mediator demand-supply gap). Model 2 (full Fed-path mediation) supported iff median $A_0^a \geq 0.70$ AND no primary family residual significant. Otherwise Model 1.

7 Null competitor horse races

Per Manual §7: structural-origin model (LP-IV with u_t^D, u_t^S) vs four null competitors:

1. Headline-only PCE inflation innovation (Fama-Schwert 1977 baseline).
2. Core/energy bivariate VAR(12) (Fang-Liu-Roussanov 2026).
3. 36-month rolling stock-bond correlation regime indicator (Bonelli-Palazzo-Yamarthy 2025).
4. Fed-path-only mediator m_t (Bernanke-Gertler-Watson 1997).

Pass threshold: family-median incremental $R^2 \geq 0.01$ AND ≥ 1 asset with HAC Wald p < 0.05 on joint $\{u^D, u^S\}$ in augmented model.

Table 5: Fed-channel mediator absorption A_0^a (Manual §6.3)

Asset	$\hat{\delta}_{with}$	$\hat{\delta}_{without}$	A_0^a	Status
UST2Y_NOM	1.337	1.250	-0.069	ok
UST10Y_NOM	-4.276	-4.381	0.024	ok
TIPS5Y_REAL	8.409	8.606	0.023	ok
TIPS10Y_REAL	3.566	3.730	0.044	ok
BEI5Y	-9.777	-9.798	0.002	ok
BEI10Y	-5.553	-5.617	0.012	ok
DKW10Y_IRP	-0.517	-0.523	0.011	ok
EQ_MKT_VW	-0.413	-0.383	-0.079	ok
EQ_CYCLICAL_12IND	-0.065	-0.036	-0.806	ok
EQ_DEFENSIVE_12IND	-0.688	-0.662	-0.038	ok
IG_OAS	—	—	—	insufficient_n
HY_OAS	—	—	—	insufficient_n
USD_BROAD	0.376	0.377	0.004	ok
CMDTY_BROAD	-4.459	-4.503	0.010	ok
CMDTY_ENERGY	-6.622	-6.674	0.008	ok
Median A_0^a across primary	—	—	**0.008**	—
Threshold for Model 2	—	—	0.70	—

Notes: $A_0^a = 1 - |\widetilde{\delta}_0^a|/|\delta_0^a|$ where $\widetilde{\delta}$ is the with-mediator gap and δ is the without-mediator gap. $A = 0$ means mediator absorbs none; $A = 1$ means full absorption. Median $A =$:.3f
bf is far below the 0.70 threshold for Model 2 (full Fed-path mediation), supporting Model 1 (direct structural-origin pricing on top of Fed path).

8 Robustness

8.1 R1: 8-variable VAR(12) innovations

8.2 R2: $h = 6$ horizon (BH FDR appendix)

9 Plan B decision tree application

Per PAP §9 / Manual §13. Booleans computed from primary results.

10 2025 holdout — out-of-sample exercise (read LAST per PAP §12)

11 Discussion

11.1 Reading the disciplined null

The headline result, H1c failure on both PAP-locked criteria, is informative rather than uninterpretable. The literature on stagflation-risk pricing (Pflueger 2025; Cieslak and Pflueger 2023) and on demand-versus-supply inflation decomposition (Shapiro 2022; Bonelli, Palazzo, and Yamarthy 2025) leaves open a strong cross-asset prediction: if investors price inflation *by source*, then a Fama-MacBeth pricing of the structural exposures should generate $\lambda_D \neq \lambda_S$ with a consistent sign

Table 6: Null competitor horse races: incremental R^2 of structural-origin model vs each competitor (median across family; Wald p<0.05 count in parens)

Family	N	vs Headline-only	vs Core/Energy	vs Stock-bond regime	vs Fed-path-only
rates_tips	7	0.0017 (0)	0.0014 (1)	0.0144	0.0145
equities	3	0.0027 (0)	0.0033 (0)	0.0022	0.0055
fx	1	0.0143 (1)	0.0147 (1)	0.0225	0.0272
commodities	2	0.0018 (0)	0.0041 (0)	0.0899	0.1400

Notes: Pass threshold = family-median incremental $R^2 \geq 0.01$ AND ≥ 1 asset with HAC Wald $p < 0.05$ on joint $\{u^D, u^S\}$ in augmented model. Competitor 1 = univariate AR(12) on PCE headline. Competitor 2 = bivariate VAR(12) on core/energy PCE. Competitor 3 = 36-month rolling stock-bond correlation regime indicator (lagged). Competitor 4 = mediator m_t alone. "insufficient_n" assets excluded.

Table 7: R1 robustness: 8-variable VAR(12) innovations, primary LP-IV at $h = 0$

Asset	$\hat{\beta}_D$	$\hat{\beta}_S$	$\hat{\delta}$	Status
`UST2Y_NOM`	1.802	0.645	1.157	ok
`UST10Y_NOM`	2.273	5.960	-3.687	ok
`TIPS5Y_REAL`	-0.598	-7.854	7.256	ok
`TIPS10Y_REAL`	0.613	-2.466	3.078	ok
`BEI5Y`	4.064	12.496	-8.432	ok
`BEI10Y`	3.730	8.516	-4.786	ok
`DKW10Y_IRP`	0.097	0.543	-0.446	ok
`EQ_MKT_VW`	0.389	0.745	-0.356	ok
`EQ_CYCLICAL_12IND`	0.447	0.503	-0.055	ok
`EQ_DEFENSIVE_12IND`	0.042	0.635	-0.593	ok
`IG_OAS`	—	—	—	insufficient_n
`HY_OAS`	—	—	—	insufficient_n
`USD_BROAD`	-0.145	-0.469	0.324	ok
`CMDTY_BROAD`	2.096	5.941	-3.845	ok
`CMDTY_ENERGY`	3.240	8.950	-5.710	ok

across asset families. Our test rejects that joint hypothesis under the locked specification. The corrected per-asset effective-sample stationary block bootstrap ($N = 5000$) yields $\hat{\delta}_{H1c} = -0.00165$ with two-sided $p = 0.060$, with the additional damning fact that the point estimate has the opposite sign from the PAP-locked +0.30 prior.

The combination of wrong-sign and $p > 0.05$ matters in a way that wrong-sign-with-significance or correct-sign-without-significance does not. Wrong-sign-with-significance might license a structural reframe (along the lines of Layer-2 fix-don't-kill in our workflow); correct-sign-without-significance might license a power-related discussion of sample size. Wrong-sign-without-significance is what the audit decision treats as the unambiguous failure mode, and we adopt that interpretation here.

11.2 What survives, and what does not

Among the five family-level W^+ tests, only the commodity family achieves Romano-Wolf adjusted significance, with six of six asset-horizon cells correctly signed. This is the strongest pattern in

Table 8: R2 robustness: $h = 6$ horizon, primary innovations + Känzig IV (BH FDR appendix)

Asset	$\hat{\beta}_D$	$\hat{\beta}_S$	$\hat{\delta}$	Status
UST2Y_NOM	5.515	-3.846	9.361	ok
UST10Y_NOM	2.056	7.064	-5.008	ok
TIPS5Y_REAL	7.677	4.814	2.863	ok
TIPS10Y_REAL	6.578	3.785	2.794	ok
BEI5Y	-4.837	-0.703	-4.134	ok
BEI10Y	-2.703	0.166	-2.869	ok
DKW10Y_IRP	0.125	0.838	-0.713	ok
EQ_MKT_VW	-0.178	-0.647	0.469	ok
EQ_CYCLICAL_12IND	-0.193	-0.843	0.650	ok
EQ_DEFENSIVE_12IND	-0.160	-0.537	0.377	ok
IG_OAS	—	—	—	insufficient_n
HY_OAS	—	—	—	insufficient_n
USD_BROAD	-0.138	-0.105	-0.033	ok
CMDTY_BROAD	2.415	5.296	-2.880	ok
CMDTY_ENERGY	3.759	7.892	-4.133	ok

the data and is consistent with the supply-shock pass-through channel for commodities embedded in the Känzig (2021) instrument. The rates-TIPS family achieves a positive W^+ statistic but no Romano-Wolf adjusted significance; the equity family, the FX family, and the credit family do not pass either threshold (with credit constrained by the ICE BofA OAS data-truncation disclosure documented in Section 2).

Among the wrong-sign disclosures, the equity-leg invalidation at $h = 0$ is the more theoretically consequential. Equity returns to demand-driven inflation u_t^D should, under most stagflation-risk-pricing models, exhibit a sign that aligns with the standard discount-rate-versus-cash-flow trade-off; the observed reversal undermines that simple version of the stock-bond beta-by-source story. The `UST10Y_NOM` wrong-sign is more circumscribed: a single asset-horizon cell, plausibly attributable to supply-side innovation effects dominating at the long nominal end. Both wrong-sign findings are reported as observed, without re-framing.

The Fed-channel mediator absorption test rejects Model 2 (full Fed-path mediation) decisively: median $A_0^a = 0.008$, far below the 0.70 threshold from Manual §6.3. Whatever pricing of structural-origin inflation exists in the cross-section is not absorbed by the Fed-path mediator m_t. This is a methodologically constructive finding: it forecloses a class of explanations under which any apparent structural-origin pricing would be a Fed-path artifact.

11.3 Reading the null competitor horse races

The four pre-registered null competitors (headline-only PCE inflation, core/energy bivariate VAR(12), 36-month rolling stock-bond correlation regime, and Fed-path-only mediator) are useful for what they rule out, not for what they support. None of the four single competitors generates a family-median incremental $R^2 \geq 0.01$ with at least one asset-level Wald rejection in the augmented model. The disciplined-null interpretation therefore extends to: *neither* our structural-origin LP-IV *nor* the four most plausible competitors successfully prices the cross-asset surface in this sample. This is a non-trivial joint statement; it constrains the menu of theories that can survive simultaneous cross-asset and pre-registered scrutiny.

Table 9: Plan B decision tree application (PAP §9 / Manual §13)

Indicator	Value
`PRIMARY_SUCCESS` (any family)	True
`CENTRAL_H1C_SUCCESS` (two-sided $p < 0.05$)	False
`FED_MODEL2_SUPPORT` (median $A \geq 0.70$)	False
Median mediator absorption A	:.4f
$\hat{\delta}_{H1c} = \hat{\lambda}_D - \hat{\lambda}_S$	:.5f
H1c two-sided p (boot N=5000 corrected)	:.4f
Family pass:	
`rates_tips`: FAIL	
`equities`: FAIL	
`credit`: FAIL	
`fx`: FAIL	
`commodities`: PASS	
Decision	**Asset-response map with H1c failure; abstract must state**
Plan B path invoked	neither

Notes: Per PAP §9 / Manual §13. Bootstrap N=5000 per-asset effective-sample (Patch 1 corrected per Round-6 audit); never resamples pre-availability NA periods.

11.4 Boundary conditions and what we cannot conclude

Three boundary conditions limit the inferences a reader can draw from the failure. First, the credit-family loss is a data-availability constraint, not a theoretical statement: the ICE BofA OAS truncation disclosed in Section 2 reduces the credit-family effective sample to $T = 8$ months, which our PAP-locked $T \geq 120$ rule excludes from primary tests. We did not silently substitute Moody's BAA-AAA, and we report the family-level loss explicitly. Second, the Shapiro (2022) decomposition is one of several available decompositions of inflation into demand and supply components; alternative decompositions (e.g., Eickmeier and Hofmann's 2022 BVAR-based decomposition) could produce different first-stage innovations. We do not test alternative decompositions in this paper. Third, the Round-6 external audit converted what would otherwise have been a Plan B reframe into a disciplined null report; under a less rigorous reporting standard, the wrong-sign $h = 0$ cells could be re-described as "regime contrast" or "directionally consistent." We have explicitly avoided that path.

11.5 Why a disciplined null result advances the literature

A null result reported under a strict pre-registration discipline narrows the parameter space of viable theories more credibly than a marginal positive result reported under flexible specifications. The asset-pricing literature has accumulated a stock of marginal-significance findings whose replicability is uncertain (cf. Hou-Xue-Zhang 2020; Harvey-Liu-Zhu 2016). Our paper is consistent with the recent move toward pre-registered tests in finance, and it demonstrates that the cost of pre-registration is largely a cost on *exaggeration*, not on contribution. We were able to publish the failure (the central H1c hypothesis is rejected, the equity leg is invalidated, one Treasury asset shows wrong-sign) without re-purposing the data to manufacture an alternative positive headline. Future work that relies on different inflation decompositions, different identifications of the supply

Table 10: 2025 holdout: Shapiro D/S innovations from locked Sample-A VAR coefficients (out-of-sample, never pooled into primary)

Month	u^D_{raw}	u^S_{raw}	u^D_{std}	u^S_{std}
2025-01	-0.444	2.013	-0.363	1.299
2025-02	1.316	0.863	1.076	0.557
2025-03	-0.964	-0.942	-0.788	-0.607
2025-04	-0.719	-0.613	-0.588	-0.395
2025-05	-0.460	0.190	-0.376	0.123
2025-06	0.824	-1.048	0.673	-0.676
2025-07	-0.498	-0.238	-0.407	-0.154
2025-08	0.642	-1.597	0.524	-1.030
2025-09	0.505	-0.299	0.413	-0.193
2025-10	-0.891	-1.151	-0.728	-0.743
2025-11	-0.202	-0.161	-0.165	-0.104
2025-12	-0.221	1.106	-0.181	0.713

Notes: Innovations computed as one-step-ahead forecast errors using bivariate VAR(12) coefficients estimated on Sample A (1974:01–2024:12). Standardized using $\sigma_{u_D}, \sigma_{u_S}$ from in-sample manifest. 2025 holdout LP-IV against the locked asset menu is documented in the audit packet but not estimated here because the 12-month sample is below the $T < 60$ minimum threshold; standardized innovations are reported for descriptive purposes.

leg, or different cross-asset coverage can build on this map without inheriting our specific failure modes.

12 Conclusion

We test, under strict pre-registration and Round-6 external audit, whether structural-origin inflation (demand-driven versus supply-driven) is priced differently across a 15-asset × 4-horizon U.S. cross-asset map. The central cross-sectional pricing hypothesis H1c, which predicts $\lambda_D - \lambda_S > 0$, fails on both PAP-locked criteria: the corrected per-asset effective-sample stationary block bootstrap ($N = 5000$) yields $\hat{\delta}_{H1c} = -0.00165$ (sign opposite the prior) with two-sided $p = 0.060$. Two PAP-locked sign predictions are violated at $h = 0$: the equity leg is invalidated, and `UST10Y_NOM` shows wrong-sign supply-leg response. Among the five family-level Romano-Wolf adjusted tests, only the commodity family clears significance, with six-of-six correctly signed asset-horizon cells. The Fed-channel mediator absorption test rejects Model 2 decisively. The four pre-registered null competitors do not outperform the structural-origin LP-IV either. The contribution is therefore a disciplined null cross-asset map of structural-origin inflation pricing under a pre-registered specification with explicit kill-shot triggers, transparent wrong-sign disclosures, and audited bootstrap inference that respects per-asset effective samples. The data and replication package are available in the public repository accompanying this paper.

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